

Raghav Kataria

Waco, Texas – 76706

in raghav-kataria9 •  Raghav Kataria

Summary

- Bioinformatics and computational biology researcher with expertise in integrating molecular biology, computational methods, and big data to derive scientific discovery and solve complex biological problems.
- Strong scientific communication skills, demonstrated by the publication of 11 research articles in high-impact, peer-reviewed journals.
- Experienced in database management, scientific writing, and interdisciplinary collaboration, with a track record of working across disciplines to develop and implement bioinformatics solutions, and adept at managing large datasets.
- Proficient in NGS data analysis, sequence analysis, network biology, and workflow automation. Experienced in analyzing various sequencing data, as well as developing automated pipelines for efficient data processing and analysis.
- Strong problem-solving and critical thinking skills, with a proven ability to analyze large-scale and complex biological datasets to derive meaningful insights.
- Inquisitive and eager to learn, with a passion for exploring new technologies, methodologies, and innovative approaches in bioinformatics. Enthusiastic about staying up to date with advancements in computational biology to enhance research capabilities

Education

Utah State University

Ph.D. in Plant Science (Specialization: Bioinformatics and Computational Biology)

GPA: 3.81/4.0

Logan, UT, USA

2020–2026

Assam Agricultural University

Masters in Agricultural Biotechnology

GPA: 8.30/10.0 (First Class with Distinction)

Assam, India

2017–2019

Punjab Agricultural University

Bachelors (Hons.) in Biotechnology

GPA: 7.63/10.0

Punjab, India

2013–2017

Research Experience

Baylor University

Postdoctoral Research Associate

Waco, TX, USA

April 2026–Present

- Applying bioinformatics and statistical approaches to study long-read metagenomics and microbial genomics using PacBio HiFi sequencing data.
- Developing and managing computational pipelines for processing and analyzing large-scale HiFi metagenomic datasets, from raw reads to genome-resolved microbial community profiles.

Utah State University

Graduate Research Assistant

Logan, UT, USA

September 2020–March 2026

- Implementing bioinformatics and computational biology approaches to study host-pathogen protein-protein interactions across species.
- Performing different kinds of NGS data analysis (RNA-Seq, Methylome, De novo transcriptome, etc.).
- Developing databases for various host-pathogen systems.

Utah State University

Grant Writing

Logan, UT, USA

May–August 2023

- Contributed to National Science Foundation (NSF) and United States Department of Agriculture (USDA) grant proposals.
- Focused on species-specific protein-protein interaction networks and the development of a comprehensive protein interaction database with functional annotations of the proteins.

Punjab Agricultural University

Senior Research Fellow

Punjab, India

January–August 2020

- Conducted expression analysis experiments on cotton pests to understand the molecular mechanisms.

Assam Agricultural University

Master's Thesis

Assam, India

August 2017–August 2019

- **Thesis Title:** In vitro regeneration of *Musa spp.* cultivars Malbhog and Bhimkol, and assessment of genetic fidelity of regenerated plantlets using molecular markers.

Punjab Agricultural University

In-house Training

Punjab, India

January–June 2017

- Conducted computational mining of miRNA and their targets in *Pisum sativum*.

Teaching Experience

Utah State University

Teaching Assistant

Logan, UT, USA

Fall 2023, 2024, & 2025

- Assisted in teaching *Bioinformatics and Big Data Mining* course, guiding students in computational methods and data analysis.

Utah State University

Student Training

Logan, UT, USA

2022–2025

- Mentored 5 students (Undergraduate and Graduate) in bioinformatics fundamentals, including Linux command-line, various tools and next-generation sequencing (NGS) data analysis.

Skills

Bioinformatics: Proficient in sequence analysis; NGS data analyses (transcriptome, single-cell RNA-Seq, de novo transcriptome, microbiome (16S & ITS), metagenome); long-read metagenomics using PacBio HiFi data; molecular docking (AutoDock Vina); and molecular dynamics simulations (GROMACS)

Programming: Skilled in Python and R for bioinformatics and data analysis applications; expertise in Linux (Bash scripting); Jupyter Notebook; experience with databases (SQL and MongoDB); adept at managing computational environments with Conda; workflow automation; Nextflow (beginner); and version control (Git and GitHub) for collaborative coding and project tracking

High-Performance Computing (HPC): Experience with cluster computing, SLURM workload management, and optimizing bioinformatics pipelines for large-scale data analysis

Computational Tools: HMMER, BLAST, ClustalW, DESeq2, edgeR, STAR, HISAT2, featureCounts, Bismark, methylkit, BUSCO, OrthoFinder, QIIME2, Kraken2, Bioconductor, Seurat, and others

Visualization & Reporting: Experienced in data visualization and reporting using ggplot2, Matplotlib, and Seaborn for statistical graphics; Cytoscape and NetworkX for network visualization; proficient in R Markdown and LaTeX for creating publication-quality figures and reproducible research reports

Molecular Biology Techniques: Experienced in DNA/RNA extraction, gel electrophoresis, PCR, qPCR, and DNA quantification

Writing & Communication: Experienced in scientific manuscript preparation, grant proposals, and technical documentation; strong ability to convey complex results clearly to interdisciplinary audiences

Organizational & Leadership: Highly organized and detail-oriented; experienced in leading teams, coordinating research projects, multitasking efficiently, meeting deadlines, and maintaining data integrity

Publications

Research Articles Published

1. **Kataria, R.**, Duhan, N., & Kaundal, R. (2024). Navigating the human-monkeypox virus interactome: HuPoxNET atlas reveals functional insights. *Frontiers in Microbiology*, 15, 1399555. <https://doi.org/10.3389/fmicb.2024.1399555> [IF: 4.0]
2. Hauck, K., **Kataria, R.**, Guevara-Barrientos, D., & Kaundal, R. (2024). HuCoVaria: a Pan-Viral SARS-CoV-2 Human Protein-Protein Interaction Atlas. *Database: The Journal of Biological Databases*

and Curation. [IF: 3.4] [Accepted; In Press]

3. **Kataria, R.**, Kaur, S., & Kaundal, R. (2023). Deciphering the complete human-monkeypox virus interactome: Identifying immune responses and potential drug targets. *Frontiers in Immunology*, March, 1–13. <https://doi.org/10.3389/fimmu.2023.1116988> [IF: 8.786]
4. **Kataria, R.** & Kaundal, R. (2022). TRustDB: A comprehensive bioinformatics resource for understanding the complete Wheat-Stem rust host-pathogen interactome. *Database: The Journal of Biological Databases and Curation*, 2022, 1–9. <https://doi.org/10.1093/database/baac068> [IF: 4.462]
5. Duhan, N., **Kataria, R.**, & Kaundal, R. (2022). TriticKBdb: A Functional Annotation Resource for Deciphering the Complete Interaction Networks in Wheat-Karnal Bunt Pathosystem. *International Journal of Molecular Sciences*, 23(13). <https://doi.org/10.3390/ijms23137455> [IF: 6.208]
6. **Kataria, R.** & Kaundal, R. (2022). Deciphering the Crosstalk Mechanisms of Wheat-Stem Rust Pathosystem: Genome-Scale Prediction Unravels Novel Host Targets. *Frontiers in Plant Science*, 13. <https://doi.org/10.3389/fpls.2022.895480> [IF: 5.753]
7. **Kataria, R.** & Kaundal, R. (2022). WeCoNET: a host–pathogen interactome database for deciphering crucial molecular networks of wheat-common bunt cross-talk mechanisms. *Plant Methods*, 18(1), 1–11. <https://doi.org/10.1186/s13007-022-00897-9> [IF: 5.139]
8. **Kataria, R.**, Duhan, N., & Kaundal, R. (2022). Computational Systems Biology of Alfalfa – Bacterial Blight Host-Pathogen Interactions: Uncovering the Complex Molecular Networks for Developing Durable Disease Resistant Crop. *Frontiers in Plant Science*, 12: 807354. <https://doi.org/10.3389/fpls.2021.807354> [IF: 5.753]
9. **Kataria, R.** & Kaundal, R. (2022). Deciphering the Host-Pathogen Interactome of the Wheat-Common Bunt System: A Step towards Enhanced Resilience in Next Generation Wheat. *International Journal of Molecular Sciences*, 23(5): 2589. <https://doi.org/10.3390/ijms23052589> [IF: 5.923]
10. Hu, X.-L., Zhang, J., Kaundal, R., **Kataria, R.**, Labbé, J. L., Mitchell, J. C., Tschaplinski, T. J., Tuskan, G. A., Cheng, Z.-M. (Max), & Yang, X. (2022). Diversity and conservation of plant small secreted proteins associated with arbuscular mycorrhizal symbiosis. *Horticulture Research*, 1–13. <https://doi.org/10.1093/hr/uhac043><https://doi.org/10.1093/hr/uhac043> [IF: 6.07]
11. **Kataria, R.** & Kaundal, R. (2021). alfaNET: A Database of Alfalfa-Bacterial Stem Blight Protein–Protein Interactions Revealing the Molecular Features of the Disease-causing Bacteria. *International Journal of Molecular Sciences*, 22(15): 8342. <https://doi.org/10.3390/ijms22158342> [IF: 5.923]

Research Articles (Under review).....

1. **Kataria, R.**, McGarry, T., Costello, M., Wiegert, S., Otten, A., Portik, D., Kearney, C., Mellor, J., Manery, M., & Minich, J. (2026). Long-read tagmentation unlocks scalable, cost-effective, HiFi metagenomics from low-input DNA. *Nature Microbiology*. [IF: 18.7]
2. **Kataria, R.**, Jayee, B., & Kaundal, R. (2026). Identification of Broad-Spectrum Natural Antifungals Against Soybean Diseases via Multi-Target Docking and Molecular Dynamics Simulations. *Pest Management Science*. [IF: 4.2]
3. **Kataria, R.** & Kaundal, R. (2026). HPIpy: A Multi-Model Toolbox for Predicting Host-Pathogen Protein-Protein Interactions using Computational Models. *Bioinformatics*. [IF: 5.5]

Research Articles (Manuscript prepared).....

1. Azeez, I., Norton, J., Gudla, L., Atoloye, I., Jacobson, A., Reeve, J., Kaundal, R., & **Kataria, R.** (2025). Single Compost Application Affects Microbial Communities in an Organic Dryland Wheat System. *Soil Biology and Biochemistry*. [IF: 10.3]
2. **Kataria, R.**, Lalit, B., & Kaundal, R. (2025). Systems-Level Profiling of Wheat-Stripe Rust Interactions: Functional Enrichment and Insights into Host-Pathogen Responses. *Plant Stress*. [IF: 6.8]
3. Larson, R., **Kataria, R.**, & Kaundal, R. (2025). myDockDB: A Database for Docking Natural Compounds against Multiple Myeloma Proteins. *Journal of Molecular Graphics and Modelling*. [IF: 2.7]
4. **Kataria, R.**, Kaur, S., & Kaundal, R. (2024). Decoding the *Triticum-Karnal Bunt* Protein Interactome: Unraveling Complex Molecular Networks and Novel Disease Targets in Wheat. *Current Plant Biology*. [IF: 5.4]
5. Duhan, N., **Kataria, R.**, & Kaundal, R. (2024). HPInet: An Atlas for Navigating the Landscape of Plant-Pathogen Protein-Protein Interactions. *Briefings in Bioinformatics*. [IF: 9.5]

6. **Kataria, R.**, Duhan, N., & Kaundal, R. (2024). Decoding Intracellular Dynamics: Deep Learning-based Exploration of Subcellular Localization of Legume Crop Proteins. *Artificial Intelligence in Agriculture*. [IF: 5.7]
7. **Kataria, R.** & Kaundal, R. (2024). Molecular Defense Against Fungal Threat: A Comprehensive Study of Wheat-Dwarf Bunt Pathosystem. *Plant Physiology*. [IF: 7.4]

Review Articles (Published).....

1. Raghavendra, N. R. & **Kataria, R.** (2026). Breeding for iron rich maize (*Zea mays* L.) progress, prospects and challenges. *Journal of Crop Science and Biotechnology*. <https://doi.org/10.1007/s12892-025-00336-w> [IF: 1.5]

Review Articles (Manuscript prepared).....

1. Kaur, S., Seem, K., **Kataria, R.**, Kumar, S., & Kaundal, R. (2024). High-throughput Chromosome Conformation Capture (Hi-C): Relevance in Understanding 3D Genome Architecture and Assembly. *Frontiers in Cell and Developmental Biology*. [IF: 4.6]

Book Chapters (Published).....

1. Paul, S., Khataniar, L., Sharma, K., **Kataria, R.**, & Enghipi, J. (2026). Effect of Integrated Nutrient Management (INM) on forage crop under Medium Land Situation. *Taylor & Francis*.

Book Chapters (Under review).....

1. **Kataria, R.** & Kaundal, R. (2025). Computational Prediction of Host-Microbe Protein-Protein Interactions: From Data to Discovery. *Springer Protocols*.

Conferences Attended and Presented

1. Kataria, R.. (2026). Long-read tagmentation unlocks scalable, cost-effective, HiFi metagenomics from low-input DNA. **Baylor University Department of Biology Graduate Program Retreat**; August 13. Oral presentation.
2. Kataria, R., Jayee, B., and Kaundal, R*. (2025). Structure-Based Virtual Screening and Molecular Dynamics Evaluation of Natural Compounds Targeting Fungal Proteins in Soybean Diseases. **Utah State University Hansen Retreat**; October 25. Poster presentation.
3. Kataria, R., and Kaundal, R*. (2025). HPIpy: A Multi-Model Python-based Toolbox for Predicting Host-Pathogen Protein-Protein Interactions using Diverse Computational Models. **MidSouth Computational Biology and Bioinformatics Society (MCBIOS) Conference**; March 27-29. Oral presentation.
4. Kataria, R., and Kaundal, R*. (2024). HPIpy: Decoding Host-Microbe Crosstalk with Computational Homology Models for Protein Interaction Prediction. **Utah State University Student Research Symposium**; April 9. Oral presentation.
5. Kataria, R., and Kaundal, R*. (2023). WeCoNET: A Host-Pathogen Interactome Database for Deciphering Crucial Molecular Networks of Wheat-Common Bunt Cross-Talk Mechanisms. **Plant & Animal Genome Conference (PAG 30)**; January 13-18. Oral presentation in "Digital Tools and Resources" section.
6. Kataria, R., Kaur, S., and Kaundal, R*. (2023). Deciphering the Complete Human-Monkeypox Virus Interactome: Identifying Immune Responses and Potential Drug Targets. **Plant & Animal Genome Conference (PAG 30)**; January 13-18. Poster presentation.
7. Kataria, R., and Kaundal, R*. (2022). Genome-scale Protein Interactome: Unraveling Novel Host Targets in Wheat-Stem Rust Pathosystem. **Intelligent Systems for Molecular Biology (ISMB) 2022**; Virtual Conference; July 10-14. Oral presentation.
8. Kataria, R., and Kaundal, R*. (2021). Computational systems biology of Wheat - Common bunt protein-protein interactions: Uncovering the complex molecular networks in an important host-pathogen interaction system. **2021 International Society for Molecular Plant-Microbe Interactions Congress**; Virtual Conference; December 1-2. Oral presentation.
9. Kataria, R., and Kaundal, R*. (2021). Translating -omics big data: comprehensive understanding of host-pathogen interactions to control common bunt in wheat using computational approaches. **2021 ASA-CSSA-SSSA International Annual Meeting**; November 7-10. Oral presentation.
10. Kataria, R., Duhan, N., and Kaundal, R*. (2021). Computational systems biology of Alfalfa - *Pseudomonas syringae* protein-protein interactions: Uncovering the complex molecular networks of host-pathogen interactions to control bacterial blight in Alfalfa. **2021 ASA-CSSA-SSSA Interna-**

tional Annual Meeting; November 7-10. Oral presentation.

11. *Kataria, R., and Kaundal, R*. (2021). alfaNET: A Database of Alfalfa-Bacterial Stem Blight Protein-Protein Interactions Revealing the Molecular Features of the Disease-causing Bacteria. 2021 International Society for Molecular Plant-Microbe Interactions Congress*; Virtual Conference; September 15-16. Oral presentation.
12. *Kataria, R., Duhan, N., and Kaundal, R*. (2021). Translating -omics big data: comprehensive understanding of host-pathogen interactions to control bacterial blight in Alfalfa using computational approaches. 2021 International Society for Molecular Plant-Microbe Interactions Congress*; Virtual Conference; July 12-13. Oral presentation.
13. *Kataria, R., Duhan, N., and Kaundal, R*. (2021). Translating -omics big data: comprehensive understanding of host-pathogen interactions to control bacterial blight in Alfalfa using computational approaches. Utah State University Student Research Symposium*; Virtual Symposium; April 14-15. Oral presentation.

Trainings Attended

1. Attended **Hands-on Training in Molecular Modeling & Molecular Dynamics Simulation**, organized by Distributed Information Centre (DIC), Department of Agricultural Biotechnology, Assam Agricultural University, Jorhat, Assam; February 25-27, 2019.
2. Attended **ABLE-BEST INDIA 2018 Entrepreneurship Workshop**, organized by Department of Biotechnology and ABLE (Association of Biotechnology Led Enterprises); December 17-21.

Talks Delivered

1. Delivered a talk on **"Uncovering the Hidden Connections: Deciphering the complete Protein-Protein Interactome in Wheat-Stem Rust Pathosystem"** in the Department of Plants, Soils, & Climate, Utah State University, Logan, USA; April, 2023.
2. Delivered a talk on **"Translating -omics big data: A comprehensive understanding of host-pathogen interactions to control bacterial blight in Alfalfa"** in the Department of Plants, Soils, & Climate, Utah State University, Logan, USA; October, 2021.
3. Presented a credit seminar on **"Chloroplast-to-nucleus Retrograde Signalling"** in the Department of Agricultural Biotechnology, Assam Agricultural University, Jorhat, Assam; November, 2018.

Activities, Participations & Accolades

Academic Roles.....

1. Serving as a reviewer for **Scientific Reports** (Impact Factor: 3.9) (September 2025 – Present).
2. Serving as a reviewer for **BMC Plant Biology** (Impact Factor: 4.8) (July 2025 – Present).
3. Serving as a reviewer for **Frontiers in Plant Science** (Impact Factor: 4.1) (May 2025 – Present).
4. Serving as a reviewer for **Frontiers in Cellular and Infection Microbiology** (Impact Factor: 4.6) (April 2025 – Present).
5. Serving as a reviewer for **Oxford Academic Database** (Impact Factor: 3.4) (January 2025 – Present).
6. Serving as a reviewer for **MDPI Plants** (Impact Factor: 4.0) (December 2024 – Present).
7. Serving as a reviewer for **Computers in Biology and Medicine** (Impact Factor: 7.7) (December 2023 – Present).
8. Serving as a reviewer for **Journal of Advanced Research** (Impact Factor: 13.0) (June 2023 – Present).
9. Serving as a reviewer for **Computational and Structural Biotechnology Journal** (Impact Factor: 6.155) (April 2022 – Present).

Scholarships.....

1. Awarded with the **"Ambassador Ardeshtir Zahedi International Endowment Scholarship"** at Utah State University, Logan (2025 – 2026).
2. Awarded with the **"Apogee Instruments – Campbell Scientific Graduate Fellowship"** at Utah State University, Logan (2024 – 2025).
3. Awarded with the **"Bertrand D. Tanner/Campbell Scientific Graduate Fellowship"** at Utah State University, Logan (2023 – 2024).
4. Awarded with the **"Apogee Instruments – Campbell Scientific Graduate Fellowship"** at Utah

State University, Logan (2022 – 2023).

5. Awarded with the “**AGRI Ambassador Ardeshir Zahedi International Endowment Scholarship**” at Utah State University, Logan (2021 – 2022).
6. Awarded with **ICAR-NTS fellowship** for P.G. Programme at Assam Agricultural University, Jorhat, India (2017 – 2019).

Other Roles.....

1. Served as the **President**, International Student Council (ISC) at Utah State University, Logan, Utah, USA (May 2023 – April 2024).
 - Led an 8-member executive council, served as the official spokesperson
 - Organized events to enhance international student engagement and foster an inclusive environment
2. Served as **Public Relations & Marketing Vice President**, International Student Council (ISC) at Utah State University, Logan, USA (January 2022 – April 2023).
 - Designed marketing materials and managed event promotions across multiple platforms
 - Built and maintained relationships with on-campus and off-campus organizations